Sensors Problem Set — Complete Edition

The Full Superset: All 45 Problems from Sets 1 and 2

Conceptual, analytical, and Python problems with full solutions, organized by topic

Introduction to Electrical Engineering

August 9, 2026

Unless otherwise stated: $T\,[\mathrm{K}] = T\,[^\circ\mathrm{C}] + 273.15$; the "demo" hardware is $V_{\mathrm{in}} = V_{\mathrm{ref}} = 3.3$ V, fixed resistor $R_1 = 10\,\mathrm{k}\Omega$ at the bottom of the divider, NTC thermistor ($R_0 = 10\,\mathrm{k}\Omega$ at $25\,^\circ\mathrm{C}$, $\beta = 3950$ K) at the top, and a 10-bit ADC. Problems are grouped by topic; solutions, in matching order, begin on page 7.

Problems

A. Fundamentals, Measurement, and Calibration

Problem 1 (conceptual). A digital kitchen scale displays mass to 0.1 g but consistently reads 4.2 g high. Classify the instrument in terms of precision, accuracy, and resolution, and give one situation where it is still perfectly usable and one where it is not.

Problem 2 (conceptual). Give two examples of the observer effect: the tire-gauge example from the notes, restated in your own words, and one *electrical* example relevant to a thermistor circuit. For the electrical example, name the physical mechanism.

Problem 3 (conceptual). A manufacturer sells (a) a \$2 mass-market thermometer module and (b) a \$3,000 laboratory reference thermometer. Explain how the economics of calibration differ between the two, and describe the mechanism that lets some cheap IC sensors achieve good accuracy anyway.

Problem 4 (conceptual). The lecture recommends choosing a microcontroller with a built-in ADC rather than pairing an ADC-less microcontroller with a discrete ADC. Give two engineering reasons supporting this recommendation. Then name the third architecture on the lecture slide that avoids the analog question entirely.

Problem 5 (conceptual). Why does a linear approximation of a thermistor's R–T curve produce large errors even over the modest range 0–100 °C, and what property of the lookup-table method lets it use linear mathematics anyway?

Problem 6 (conceptual). A datasheet advertises a "16-bit temperature sensor with $\pm 2\,^\circ$C accuracy." Using the precision/accuracy/resolution vocabulary, explain what the 16 bits do and do not buy you, and compute the resolution in °C if the 16 bits span -40 to $+125\,^\circ$C.

Problem 7 (analytical). (a) Compute the LSB size, in millivolts, of the demo ADC (10-bit, 3.3 V reference). (b) Repeat for 8-bit and 12-bit configurations. (c) What count does an input of 2.00 V produce in the 10-bit configuration, and what is the maximum quantization error in millivolts?

Problem 8 (analytical). A calibration lookup table contains raw-vs-true pairs $(360, 471.7)$ and $(450, 737.1)$. (a) Estimate the true value for a raw reading of 400 by linear interpolation. (b) The underlying true response is $y = 0.00364\,x^2$; compute the exact value and the interpolation error in percent. (c) State qualitatively how the error changes if the table spacing is halved.

Problem 9 (Python). Monte Carlo calibration study: simulate 500 thermistors whose true β is drawn from $\mathcal{N}(3950, 40^2)$ and R_0 from $\mathcal{N}(10\,\mathrm{k}\Omega, (100\,\Omega)^2)$ (1% tolerance). For each, compute the temperature error at $37\,^\circ$C made by a reader that assumes the nominal constants. Report the mean and standard deviation of the error and the 95th-percentile absolute error, and state the implication for uncalibrated mass production.

B. Thermistors and the Divider Readout

Problem 10 (conceptual). Explain why an NTC thermistor placed at the *top* of a voltage divider produces an output voltage that *rises* with temperature. What happens to the direction of the output swing if you swap the thermistor to the bottom position?

Problem 11 (analytical). Using the β model, compute the demo thermistor's resistance at (a) $0\,°\mathrm{C}$, (b) $50\,°\mathrm{C}$, and (c) $100\,°\mathrm{C}$. Comment on the ratio $R(0\,°\mathrm{C})/R(100\,°\mathrm{C})$ and what it implies about choosing an ADC range.

Problem 12 (analytical). The demo thermistor measures $R = 12.0\,\mathrm{k\Omega}$. Find the temperature in °C. Before computing, state whether the answer must be above or below $25\,°\mathrm{C}$ and why.

Problem 13 (analytical). A thermistor datasheet gives Steinhart–Hart constants $A = 1.129 \times 10^{-3}$, $B = 2.341 \times 10^{-4}$, $C = 8.775 \times 10^{-8}$ (SI units). Compute the temperature for $R = 10\,\mathrm{k\Omega}$ and for $R = 5\,\mathrm{k\Omega}$.

Problem 14 (analytical). For the demo divider, compute V_{out} at $0\,°\mathrm{C}$, $25\,°\mathrm{C}$, and $50\,°\mathrm{C}$, using your resistances from Problem 11. Then verify the $25\,°\mathrm{C}$ value with the shortcut argument from the notes' Example 28.2 (no exponentials allowed).

Problem 15 (analytical). Derive the inverse of the divider equation — R as a function of V_{out} — and use it to find the thermistor resistance and temperature when the 10-bit ADC reports a count of 720.

Problem 16 (analytical). Self-heating: the demo thermistor has dissipation constant $\delta = 1.5\,\mathrm{mW/°C}$. (a) Show that the power dissipated in the thermistor is $P = V_{\mathrm{in}}^2 R/(R_1 + R)^2$ and evaluate it at $25\,°\mathrm{C}$. (b) Compute the self-heating temperature error. (c) The divider is re-designed with $R_1 = R_0 = 100\,\mathrm{k\Omega}$ (a $100\,\mathrm{k\Omega}$ thermistor). By what factor does the error shrink?

Problem 17 (analytical). Two-point β extraction: a thermistor measures $R = 10.00\,\mathrm{k\Omega}$ at $25.0\,°\mathrm{C}$ and $R = 3.588\,\mathrm{k\Omega}$ at $50.0\,°\mathrm{C}$. Compute β. Then predict R at $37.0\,°\mathrm{C}$ with your fitted model.

Problem 18 (analytical). The demo thermistor reads $R = 6.90\,\mathrm{k\Omega}$. (a) Find T. (b) Find the divider output V_{out} and the 10-bit ADC count. (c) Find the temperature error if the firmware mistakenly uses $\beta = 3977$ K instead of 3950 K.

Problem 19 (analytical). Derive, from $V_{\mathrm{out}} = V_{\mathrm{in}} R_1/(R_1 + R)$ and the β model, the divider sensitivity dV_{out}/dT at the symmetric point $R = R_1$, and evaluate it for the demo hardware at $25\,°\mathrm{C}$. Then compute the temperature resolution per 10-bit LSB there.

Problem 20 (Python). Write a function `ntc_R(T_C, R0, beta)` implementing the β model, and use it to plot $R(T)$ for the demo part over 0–100 °C on linear and semilog axes. On the semilog plot, explain the near-straightness of the curve.

Problem 21 (Python). Write `counts_to_tempC(counts)` implementing the full counts → voltage → resistance → t chain for the demo hardware, and evaluate it for counts $= [400, 512, 640, 800]$. Guard against the invalid endpoint counts 0 and 1023 and explain physically why they are invalid.

Problem 22 (Python). Temperature resolution of the digitized system: for each integer count from 200 to 900, compute the temperature; then compute the difference in temperature between adjacent counts and plot it versus temperature. Where is the system's temperature resolution best, and how does this relate to the choice $R_1 = R_0$?

C. Bridge, RTD, Sensitivity, and Noise

Problem 23 (analytical). Wheatstone bridge, equal arms $R = 350\,\Omega$, $V_s = 5$ V. (a) Derive $V_o = V_s\,\delta R/(4R + 2\delta R)$ from the general bridge equation. (b) Compute V_o exactly and by the small-signal approximation for $\delta R = 1.75\,\Omega$ (0.5%). (c) At what $\delta R/R$ does the approximation err by 1%?

Problem 24 (conceptual). Explain, with the offset argument, why a Wheatstone bridge rather than a single voltage divider is used to read a strain gauge whose full-scale resistance change is 0.2%, given that the readout chain includes a gain-1000 amplifier.

Problem 25 (analytical). pt100 RTD ($A = 3.9083 \times 10^{-3}$, $B = -5.775 \times 10^{-7}$, $C = -4.183 \times 10^{-12}$). (a) Compute R_T at 50 °C and at 200 °C. (b) Compute the sensitivity dR/dT at 50 °C and express it in %/°C. (c) Compare with the thermistor's $\approx$ 4%/°C and state one advantage each device holds over the other.

Problem 26 (analytical). A glucose sensor follows $I(C) = I_{\max}C/(K_m + C)$ with $I_{\max} = 0.30\,\mu$A and $K_m = 8$ mM. (a) Derive the sensitivity $S(C)$. (b) Evaluate S at 4 mM and at 24 mM. (c) Find the concentration at which the sensitivity has dropped to half its zero-concentration value.

Problem 27 (analytical). A sensor output is sampled ten times; deviations from the mean, in mV: $\{+3, -2, +1, +4, -3, -1, +2, -4, +1, -1\}$. (a) Compute x_{rms}. (b) How many samples must be averaged so that the standard deviation of the average falls below 0.5 mV, assuming independent noise? (c) Name two physical sources of such noise from the lecture list.

D. Chemical and Other Sensors

Problem 28 (conceptual). An ionizing smoke detector and an optical smoke detector both respond to smoke entering a chamber. For each, state the quiescent (clean-air) condition of the sensed signal and the direction of change when smoke enters, and explain the physical cause.

Problem 29 (analytical). An MQ-4 gas sensor's sensing resistance is $R_s = 18\,\mathrm{k\Omega}$ in clean air and $6\,\mathrm{k\Omega}$ at the alarm concentration. It is placed at the top of a 5 V divider with load resistor R_L at the bottom. (a) Choose R_L as the geometric mean of the two extreme resistances. (b) Compute V_{out} in clean air and at alarm. (c) Propose a digital threshold (in volts) halfway between the two, and convert it to a 10-bit count for a 5 V-referenced ADC.

E. Optical Biosensing and Pulse Oximetry

Problem 30 (analytical). Light through a 2.0 mm cuvette of a 25 mM solution is attenuated to 40% of its incident intensity. (a) Compute the absorbance and the molar absorptivity in $M^{-1}cm^{-1}$. (b) What transmittance would a 1.0 mm cuvette give? (c) What concentration in the 2.0 mm cuvette would give 10% transmittance?

Problem 31 (analytical). A mixture contains species X ($\varepsilon_X = 120\,M^{-1}cm^{-1}$, $c_X = 2$ mM) and species Y ($\varepsilon_Y = 300\,M^{-1}cm^{-1}$, $c_Y = 1.5$ mM) in a 5 mm path. Using additivity of absorbance, compute the total absorbance and the transmittance. What fraction of the absorbance is due to Y?

Problem 32 (conceptual). Explain why pulse oximetry needs (a) *two* wavelengths, (b) the *AC/DC normalization* of each channel, and (c) an *empirical* calibration curve rather than the pure Beer–Lambert prediction.

Problem 33 (analytical). A pulse oximeter measures red AC = 15 mV on DC = 600 mV, and IR AC = 22 mV on DC = 400 mV. (a) Compute R_{OS}. (b) Estimate SpO_2 with the linear calibration $SpO_2 \approx 110 - 25R_{OS}$. (c) Should this reading trigger a hypoxemia concern (threshold ~90%)?

Problem 34 (conceptual). Contrast transmission-mode and reflectance-mode pulse oximetry: sensing geometry, permissible body locations, and the best reflectance site according to the lecture's body-map data.

Problem 35 (analytical). MAX30102 bookkeeping. Each SpO_2 sample stores 3 bytes per channel (red + IR) in a 32-sample FIFO. (a) At 400 samples/s, how often must the host service the FIFO to avoid overflow, and what average I^2C data rate (bytes/s) does draining it require? (b) The temperature sensor need only be read about once per second; at 29 ms per temperature conversion, what fraction of the time is the die-temp ADC busy?

Problem 36 (conceptual). The MAX30102 contains its own 18-bit ADC, digital filtering, and ambient-light cancellation, and talks to the microcontroller over I^2C. Map this onto Lecture 13's three data-gathering architectures and explain why an integrated AFE is preferred for a PPG signal in particular (think signal amplitudes and interference).

F. Op-Amp Interface Circuits

Problem 37 (conceptual). State the three Golden Rules of op-amp circuit analysis, and explain using the open-loop relation $V_{out} = A(V_+ - V_-)$ with $A = 10^6$ why Rule 2 ("$V_+ = V_-$") holds whenever the circuit has negative feedback and the output is not saturated.

Problem 38 (analytical). For an inverting amplifier with $R_i = 4.7\,k\Omega$ and $R_f = 47\,k\Omega$: (a) find V_{out} for $V_{in} = -120$ mV; (b) find the input current; (c) redesign R_f for a gain of -33.

Problem 39 (analytical). A sensor produces 0–60 mV and the ADC wants 0–3.0 V. (a) Design a non-inverting amplifier (choose $R_i = 1\,\mathrm{k}\Omega$) for the required gain. (b) Explain why an inverting design with the same gain magnitude is awkward on a single-supply (0/3.3 V) board. (c) What gain does your circuit have if the feedback resistor accidentally opens ($R_f \to \infty$), and what happens to the output?

Problem 40 (analytical). A photodiode's full-scale signal current is $2.5\,\mu$A and the ADC spans 3.3 V. (a) Choose R_f for a transimpedance amplifier that maps full scale to $|V_{\mathrm{out}}| = 3.0$ V. (b) With ambient light adding a constant $0.8\,\mu$A, what is the output range, and does it clip? (c) Referring to the oximeter read circuit's two-stage structure, explain how the DC-subtraction DAC solves the problem in (b).

G. Signal Processing

Problem 41 (conceptual). A square wave and a sine wave have the same fundamental frequency. Describe their spectra, state which harmonics the square wave contains and how their amplitudes scale, and explain what low-pass filtering a square wave at just above its fundamental produces.

Problem 42 (analytical). A PPG is sampled at $f_s = 50$ Hz. (a) State the Nyquist frequency. (b) The heart rate is 84 BPM; give the fundamental frequency and the number of harmonics that fit below 10 Hz. (c) You must remove 50 Hz mains interference *before* the ADC. Which filter type do you use, and why can it not be done in software after sampling at this f_s?

Problem 43 (analytical). Peaks are detected in a PPG at $t = \{0.62, 1.44, 2.29, 3.10, 3.95, 4.78\}$ s. (a) Compute each beat-to-beat heart rate and the mean. (b) The FFT of the same record shows its largest peak at 0.083 Hz and its largest peak inside 0.5–4 Hz at 1.22 Hz. Which corresponds to the heart rate, what is it in BPM, and what does the 0.083 Hz peak represent?

Problem 44 (Python). Synthesize a 10 s PPG at $f_s = 100$ Hz: fundamental 1.5 Hz (90 BPM) with a $0.3\times$ second harmonic, plus baseline drift $0.8 \sin(2\pi\, 0.08t)$ and Gaussian noise of amplitude 0.2. Implement smoothing (moving average), baseline subtraction, peak detection, and report the mean beat-to-beat HR and the FFT-based HR restricted to 0.5–4 Hz. Compare both to 90 BPM.

Problem 45 (Python). Design a 3rd-order Butterworth band-pass (0.5–10 Hz) at $f_s = 100$ Hz with `scipy.signal.butter`, apply it with `filtfilt` to the raw signal of the previous problem, and show that peak detection on the filtered signal gives the same HR without an explicit baseline-subtraction step. Explain why band-passing subsumes both smoothing and drift correction.

Solutions

Solutions A. Fundamentals, Measurement, and Calibration

Solution 1. Precision: high (repeatable, 0.1 g steps). Resolution: 0.1 g. Accuracy: poor — a +4.2 g systematic offset. Usable: comparing or tracking *changes* on the same scale (e.g., "add 30 g of flour" by difference), since a constant offset cancels in differences. Not usable: absolute measurements against an external requirement (e.g., legal-for-trade weighing, or compounding medication), unless the offset is calibrated out.

Solution 2. Tire gauge: connecting the gauge bleeds a small amount of air out of the tire, so the act of measurement lowers the very pressure being measured. Electrical example: *self-heating* — the measurement current through the thermistor dissipates I^2R power, warming the bead above the ambient temperature it is supposed to report. Mechanism: Joule heating combined with the bead's finite thermal dissipation constant.

Solution 3. Mass-market: per-unit calibration cost (labor, oven time, handling) would exceed the product price, so the design must tolerate part-to-part spread; individual calibration is prohibitively expensive. Laboratory instrument: the price supports individual calibration against traceable standards, often with a calibration certificate. The escape hatch for cheap parts: **factory calibration** — the manufacturer measures each die during production and writes correction constants (a lookup table or trim values) into on-chip ROM, e.g., laser-calibrated IC humidity sensors; amortized over millions of units this is cheap.

Solution 4. Built-in ADC: (i) fewer parts, less board area, lower cost and fewer failure points; (ii) no external analog bus to route — the signal is digitized at the microcontroller, avoiding the noise, wiring, and protocol overhead of connecting a discrete converter (the lecture labels the discrete-ADC option "bad idea"). Third architecture: a **digital sensor** with its own internal ADC that connects directly to a microcontroller data port (e.g., $\mathrm{I^2C}$).

Solution 5. The thermistor's physics is exponential: $R = R_0 e^{\beta(1/T-1/T_0)}$, so R changes by a factor of ~30 over 0–100 °C; no single straight line tracks an exponential over such a span. The lookup table succeeds because it applies linearity only *piecewise*: over the short interval between adjacent table entries the curve is locally nearly straight, so chord interpolation is accurate, with error shrinking quadratically in the entry spacing.

Solution 6. The 16 bits set the *resolution*: step size $= 165\,°\mathrm{C}/2^{16} = 165/65536 = 0.0025\,°\mathrm{C}$. They buy fineness and repeatability of the reported number (precision), not closeness to the truth: accuracy remains ±2 °C, dominated by calibration, not quantization. Readings are reported to millikelvin fineness while possibly sitting 2 °C from the truth — precise, coarse-accurate. The fine resolution is still useful for detecting small *changes* and trends.

Solution 7. (a) $\Delta V = 3.3/2^{10} = 3.3/1024 = 3.22\,\mathrm{mV}$. (b) 8-bit: $3.3/256 = 12.9\,\mathrm{mV}$; 12-bit: $3.3/4096 = 0.806\,\mathrm{mV}$. (c) counts $= \lfloor 2.00/3.3 \times 1024 \rfloor = \lfloor 620.6 \rfloor = 620$; maximum quantization error is $\pm\frac{1}{2}\,\mathrm{LSB} = \pm 1.6\,\mathrm{mV}$.

Solution 8. (a) $y = 471.7 + (737.1 - 471.7)\frac{400-360}{450-360} = 471.7 + 265.4 \times 0.4444 = 589.7$. (b) Exact: $0.00364 \times 400^2 = 582.4$; error $= 7.3$, i.e., $+1.25\%$. (c) Chord error for a smooth curve scales as the *square* of the interval width, so halving the spacing cuts the error by about $4\times$.

Solution 9.

```python
import numpy as np
rng = np.random.default_rng(0)
N = 500
beta = rng.normal(3950, 40, N)
R0   = rng.normal(10e3, 100, N)
T    = 273.15 + 37.0

# each unit's actual resistance at 37 C
R = R0 * np.exp(beta * (1/T - 1/298.15))
# reader assumes nominal constants
T_read = 1.0 / (1/298.15 + np.log(R/10e3)/3950.0) - 273.15
err = T_read - 37.0
print(err.mean(), err.std(), np.percentile(np.abs(err), 95))
# -> mean ~ -0.00, std ~ 0.26, 95th pct |err| ~ 0.52  (deg C)
```

Typical result: near-zero mean error, standard deviation $\approx$ 0.25–0.3 °C, and a 95th-percentile absolute error around 0.5 °C. Implication: with 1% parts and no per-unit calibration, roughly $\pm$0.5 °C accuracy is the realistic floor — fine for a thermostat, inadequate for a clinical thermometer, which is precisely why medical-grade products pay for calibration or factory-trimmed sensors.

Solutions B. Thermistors and the Divider Readout

Solution 10. With the NTC on top and R_1 on the bottom, $V_{\text{out}} = V_{\text{in}} R_1/(R_1 + R)$. Rising temperature lowers R, shrinking the denominator, so V_{out} rises. Swapping positions gives $V_{\text{out}} = V_{\text{in}} R/(R_1 + R)$, which *falls* as temperature rises. Both work; only the sign of the slope changes.

Solution 11. $R = 10^4 \exp[3950(1/T - 1/298.15)]$: (a) T=273.15: exponent $= 3950(3.66099 - 3.35402) \times 10^{-3} = 1.2125$, $R = 33.62\,\text{k}\Omega$. (b) T=323.15: exponent $= 3950(3.09454 - 3.35402) \times 10^{-3} = -1.0249$, $R = 3.588\,\text{k}\Omega$. (c) T=373.15: exponent $= 3950(2.67989 - 3.35402) \times 10^{-3} = -2.6628$, $R = 0.698\,\text{k}\Omega$. Ratio ≈ 48: the divider must be designed so that neither extreme pushes V_{out} into the flat, information-poor ends of the ADC range; $R_1 = R_0$ centers the useful swing.

Solution 12. $R > R_0$, and for an NTC higher resistance means *colder*, so $T < 25\,°\text{C}$. $1/T = 1/298.15 + \ln(1.2)/3950 = 3.35402 \times 10^{-3} + 0.18232/3950 = 3.40017 \times 10^{-3}$, so $T = 294.10\,\text{K} = 20.95\,°\text{C} \approx 21.0\,°\text{C}$.

Solution 13. $1/T = A + B \ln R + C(\ln R)^3$. For $R = 10^4$: $\ln R = 9.2103$; $1/T = 1.129 \times 10^{-3} + 2.341 \times 10^{-4}(9.2103) + 8.775 \times 10^{-8}(9.2103)^3 = (1.129 + 2.1561 + 0.06857) \times 10^{-3} = 3.3537 \times 10^{-3}$, $T = 298.18\,\text{K} = 25.0\,°\text{C}$. For $R = 5000$: $\ln R = 8.5172$; $1/T = (1.129 + 1.9939 + 0.05422) \times 10^{-3} = 3.1771 \times 10^{-3}$, $T = 314.75\,\text{K} = 41.6\,°\text{C}$.

Solution 14. $V_{\text{out}} = 3.3R_1/(R_1 + R)$ with $R_1 = 10\,\text{k}\Omega$: $0\,°\text{C}$ ($R = 33.62\,\text{k}$): $3.3 \times 10/43.62 = 0.757\,\text{V}$; $25\,°\text{C}$ ($R = 10\,\text{k}$): $1.650\,\text{V}$; $50\,°\text{C}$ ($R = 3.588\,\text{k}$): $3.3 \times 10/13.588 = 2.429\,\text{V}$. Shortcut at

25 °C: by definition $R = R_0 = R_1$ there, so the divider is symmetric and must output exactly half the supply, $3.3/2 = 1.65\,\text{V}$ — no exponentials needed.

Solution 15. From $V_{\text{out}} = V_{\text{in}}R_1/(R_1 + R)$: cross-multiply, $V_{\text{out}}R_1 + V_{\text{out}}R = V_{\text{in}}R_1$, so $R = R_1(V_{\text{in}}/V_{\text{out}} - 1)$. Count 720: $V_{\text{out}} = 720/1024 \times 3.3 = 2.3203\,\text{V}$; $R = 10^4(3.3/2.3203 - 1) = 10^4(0.42222) = 4.222\,\text{k}\Omega$; $1/T = 3.35402\times10^{-3} + \ln(0.4222)/3950 = 3.35402\times10^{-3} - 2.1832\times10^{-4} = 3.13570\times10^{-3}$, $T = 318.91\,\text{K} = 45.8\,°\text{C}$.

Solution 16. (a) The divider current is $I = V_{\text{in}}/(R_1 + R)$ and the thermistor dissipates $P = I^2R = V_{\text{in}}^2R/(R_1 + R)^2$. At 25 °C, $P = 3.3^2 \times 10^4/(2\times10^4)^2 = 10.89\times10^4/4\times10^8 = 272\,\mu\text{W}$. (b) $\Delta T = P/\delta = 0.272/1.5 = 0.18\,°\text{C}$. (c) With all resistances scaled by 10, P scales by 1/10 ($P = V_{\text{in}}^2R/(R_1 + R)^2 \propto 1/R$ at the symmetric point), so the error drops tenfold to $\approx 0.018\,°\text{C}$. Trade-off: higher source impedance demands a higher-impedance or buffered ADC input.

Solution 17. $\beta = \dfrac{\ln(R_1/R_2)}{1/T_1 - 1/T_2} = \dfrac{\ln(10.00/3.588)}{1/298.15 - 1/323.15} = \dfrac{1.02495}{2.59478\times10^{-4}} = 3950\,\text{K}$. Prediction at 37 °C (310.15 K): $R = 10^4\exp[3950(1/310.15 - 1/298.15)] = 10^4e^{-0.5126} = 5.99\,\text{k}\Omega$.

Solution 18. (a) $1/T = 1/298.15 + \ln(0.690)/3950 = 3.35402\times10^{-3} - 9.3963\times10^{-5} = 3.26005\times10^{-3}$, $T = 306.74$ K $= 33.6\,°\text{C}$. (b) $V_{\text{out}} = 3.3\times10/(10 + 6.9) = 1.953$ V; count $= \lfloor 1.953/3.3\times1024\rfloor = 605$. (c) With $\beta = 3977$: $1/T = 3.35402\times10^{-3} - 0.371564/3977\times\ldots$ more directly $\ln(0.69)/3977 = -9.3325\times10^{-5}$, $1/T = 3.26069\times10^{-3}$, $T = 306.68\,\text{K} = 33.5\,°\text{C}$. Error $\approx -0.06\,°\text{C}$: a 0.7% β error costs only a few hundredths of a degree this close to T_0 (the error grows with distance from T_0).

Solution 19. $dV_{\text{out}}/dT = -V_{\text{in}}R_1/(R_1 + R)^2 \cdot dR/dT$ and $dR/dT = -\beta R/T^2$, so $dV_{\text{out}}/dT = V_{\text{in}}R_1\beta R/[(R_1 + R)^2T^2]$. At $R = R_1$: $dV_{\text{out}}/dT = V_{\text{in}}\beta/(4T^2) = 3.3\times3950/(4\times298.15^2) = 36.7$ mV/°C. LSB $= 3.3/1024 = 3.22$ mV, so resolution $= 3.22/36.7 = 0.088\,°\text{C}$ per LSB.

Solution 20.

```python
import numpy as np, matplotlib.pyplot as plt

def ntc_R(T_C, R0=10e3, beta=3950.0, T0_C=25.0):
    T, T0 = T_C + 273.15, T0_C + 273.15
    return R0 * np.exp(beta * (1/T - 1/T0))

T = np.linspace(0, 100, 400)
fig, (a1, a2) = plt.subplots(1, 2, figsize=(9, 3.5))
a1.plot(T, ntc_R(T)/1e3);            a1.set_ylabel("R (kOhm)")
a2.semilogy(T, ntc_R(T)/1e3)
for a in (a1, a2): a.set_xlabel("T (deg C)"); a.grid(alpha=.3)
plt.tight_layout(); plt.show()
```

On semilog axes, $\ln R = \ln R_0 + \beta(1/T - 1/T_0)$: the curve is straight in $1/T$, and since $1/T$ is itself gently curved in T over this range, the semilog plot versus T appears *nearly* straight — the visible bow is the curvature of $1/T$.

Solution 21.

```
import numpy as np
VIN = VREF = 3.3; NBITS = 10
R1 = R0 = 10e3; BETA, T0 = 3950.0, 298.15

def counts_to_tempC(counts):
    c = np.asarray(counts, float)
    if np.any((c <= 0) | (c >= 2**NBITS - 1)):
        raise ValueError("count pinned at rail - not a valid reading")
    v = c / 2**NBITS * VREF
    R = R1 * (VIN / v - 1.0)
    return 1.0 / (1.0/T0 + np.log(R/R0)/BETA) - 273.15

print(counts_to_tempC([400, 512, 640, 800]))
# -> [15.32 25.00 36.96 56.69] deg C (approx.)
```

Count 0 means $V_{\text{out}} = 0$, i.e., $R \to \infty$; count 1023 means $V_{\text{out}} \approx V_{\text{in}}$, i.e., $R \to 0$. Both correspond to temperatures outside any physical range (formally $T \to 0$ or the log diverges) and in practice indicate an open or shorted sensor — report a fault, not a temperature.

Solution 22.

```
import numpy as np, matplotlib.pyplot as plt
# reuse counts_to_tempC from the previous problem
c  = np.arange(200, 901)
Tc = counts_to_tempC(c)
dT = np.diff(Tc)
plt.plot(Tc[:-1], dT); plt.xlabel("T (deg C)")
plt.ylabel("temperature step per LSB (deg C)")
plt.grid(alpha=.3); plt.show()
print(Tc[np.argmin(dT)])   # -> minimum step near ~25 deg C
```

The per-LSB temperature step is smallest (resolution best, $\approx 0.09\,°\text{C}$) near $25\,°\text{C}$ and grows toward both ends. This is because the divider's voltage slope dV_{out}/dT peaks where $R = R_1$; choosing $R_1 = R_0$ deliberately places that peak at the temperature of greatest interest.

Solutions C. Bridge, RTD, Sensitivity, and Noise

Solution 23. (a) With $R_1 \to R + \delta R$ and all other arms R: $V_o = V_s\big[(R+\delta R)R - R \cdot R\big]/\big[(2R+\delta R)(2R)\big] = V_s\,\delta R/(4R + 2\delta R)$. (b) Exact: $V_o = 5 \times 1.75/(1400 + 3.5) = 8.75/1403.5 = 6.234$ mV; approximation: $5 \times 1.75/1400 = 6.250$ mV (0.25% high). (c) Relative error of the approximation is $(4R + 2\delta R)/(4R) - 1 = \delta R/(2R)$; setting this to 1% gives $\delta R/R = 2\%$.

Solution 24. Both circuits produce the same *signal*, $\approx V_s\,\delta R/4R$ (millivolts), but the divider's signal rides on a $V_s/2$ offset. Multiplying the divider output by 1000 would amplify the offset to $\sim 1000 \times V_s/2$ — kilovolts, far beyond any supply, so the amplifier saturates instantly. The balanced bridge outputs ≈ 0 at rest, so the gain-1000 stage amplifies only the millivolt-level change and stays within its rails.

Solution 25. (a) $50\,°\text{C}$: $R = 100[1 + 0.195415 - 5.775 \times 10^{-7}(2500)] = 100[1.195415 - 0.00144375] = 119.40\,\Omega$. $200\,°\text{C}$: $R = 100[1 + 0.78166 - 5.775 \times 10^{-7}(4 \times 10^4)] = 100[1.78166 - 0.0231] = 175.86\,\Omega$. (b) $dR/dT = R_0[A + 2BT] = 100[3.9083 \times 10^{-3} - 1.155 \times 10^{-6} \times 50] = 100(3.8506 \times 10^{-3}) =$

$0.3851\,\Omega/°\mathrm{C}$; relative: $0.3851/119.40 = 0.32\%/°\mathrm{C}$. (c) The thermistor is $\sim 12\times$ more sensitive; the RTD is far more linear, more stable over time, and usable to much higher temperatures (661 °C).

Solution 26. (a) $S(C) = dI/dC = I_{\max}K_m/(K_m + C)^2$. (b) $S(4) = 0.30 \times 8/12^2 = 2.4/144 = 16.7\,\mathrm{nA/mM}$; $S(24) = 2.4/32^2 = 2.34\,\mathrm{nA/mM}$. (c) $S(0) = I_{\max}/K_m$; half that requires $(K_m+C)^2 = 2K_m^2$, so $C = K_m(\sqrt{2} - 1) = 8 \times 0.4142 = 3.31$ mM.

Solution 27. (a) Sum of squares $= 9 + 4 + 1 + 16 + 9 + 1 + 4 + 16 + 1 + 1 = 62$; $x_{\mathrm{rms}} = \sqrt{62/10} = \sqrt{6.2} = 2.49$ mV. (b) $N \geq (2.49/0.5)^2 = 24.8 \Rightarrow N = 25$ samples. (c) Any two of: temperature fluctuations, electromagnetic interference, instability of the (bio)sensitive element or electronic circuitry, mechanical vibration, fluid-flow artifacts such as bubbles.

Solutions D. Chemical and Other Sensors

Solution 28. Ionizing: quiescently, americium-generated ions carry a small steady current from the source casing to the detector plate; smoke particles capture and neutralize the ions, so the current *decreases*. Optical: quiescently, the IR LED points off-axis from the photodiode, so almost no light is received; smoke particles scatter LED light toward the photodiode, so the received signal *increases*.

Solution 29. (a) $R_L = \sqrt{18 \times 6}\,\mathrm{k\Omega} = \sqrt{108}\,\mathrm{k\Omega} = 10.39\,\mathrm{k\Omega}$ (use $10\,\mathrm{k\Omega}$ standard value). (b) With $R_L = 10\,\mathrm{k\Omega}$: clean air $V_{\mathrm{out}} = 5 \times 10/28 = 1.786\,\mathrm{V}$; alarm $V_{\mathrm{out}} = 5 \times 10/16 = 3.125\,\mathrm{V}$. (c) Threshold $= (1.786 + 3.125)/2 = 2.455\,\mathrm{V}$; count $= \lfloor 2.455/5 \times 1024 \rfloor = \lfloor 502.9 \rfloor = 502$.

Solutions E. Optical Biosensing and Pulse Oximetry

Solution 30. (a) $A = -\log_{10}(0.40) = 0.398$; $\varepsilon = A/(c\ell) = 0.398/(0.025\,\mathrm{M}\times 0.2\,\mathrm{cm}) = 79.6\,\mathrm{M^{-1}cm^{-1}}$. (b) Halving the path halves A: $A = 0.199$, $T = 10^{-0.199} = 0.632 = 63.2\%$ (equivalently $\sqrt{0.40}$). (c) $T = 0.10 \Rightarrow A = 1.0$; $c = A/(\varepsilon\ell) = 1.0/(79.6 \times 0.2) = 0.0628\,\mathrm{M} = 62.8$ mM.

Solution 31. $A_X = 120 \times 0.002 \times 0.5 = 0.120$; $A_Y = 300 \times 0.0015 \times 0.5 = 0.225$; $A = 0.345$; $T = 10^{-0.345} = 0.452 = 45.2\%$. Fraction from Y: $0.225/0.345 = 65\%$.

Solution 32. (a) One wavelength gives one equation but there are two unknown concentrations (HbO_2 and Hb); two wavelengths with different extinction-coefficient orderings (red: Hb $\gg$ HbO_2; NIR: reversed) give two independent equations, and by absorbance additivity the ratio fixes the saturation. (b) Raw AC amplitudes depend on LED brightness, detector efficiency, skin tone, and tissue geometry; dividing each channel's AC by its own DC cancels these multiplicative factors, leaving a quantity that depends (ideally) only on arterial blood composition. (c) Beer–Lambert assumes pure absorption in a homogeneous slab; real tissue scatters light strongly, so the theoretical curve is biased and manufacturers instead fit a second-order polynomial to data from human calibration studies.

Solution 33. (a) $R_{\mathrm{OS}} = (15/600)/(22/400) = 0.0250/0.0550 = 0.4545$. (b) $SpO_2 \approx 110 - 25 \times 0.4545 = 98.6\%$. (c) No — well above the $\sim 90\%$ hypoxemia threshold.

Solution 34. Transmission mode: LEDs and photodetector on opposite sides of the tissue; light must pass through, so only thin, transilluminable sites work — fingertips and earlobes. Reflectance mode: LEDs and detector side by side on the same surface, sensing back-scattered light; usable essentially anywhere on the body. The lecture's body map shows AC amplitude highest at the forehead for both red and NIR, mid-range on the arms, weak on legs and chest — the forehead is the best reflectance site.

Solution 35. (a) FIFO fills in $32/400 = 80$ ms, so service at least every 80 ms. Data rate: $400\,\text{sps} \times 2\,\text{ch} \times 3\,\text{B} = 2400$ B/s average (plus protocol overhead). (b) One 29 ms conversion per second: duty $= 0.029/1 = 2.9\%$.

Solution 36. It is the third architecture: a *digital sensor* with internal ADC connecting directly to a data port — no analog signal ever leaves the module. For PPG this matters because the photocurrent's AC component is tiny (tens of mV equivalent after conversion, on a large DC) and would be corrupted by ambient light and interference if routed as analog across a board; digitizing at the sensor with built-in ambient-light cancellation and filtering, then sending noise-immune $\mathrm{I^2C}$ data, preserves the signal. It also bundles the LED drivers and timing that a discrete design would need several chips to replicate.

Solutions F. Op-Amp Interface Circuits

Solution 37. Rules: (1) no current into either input; (2) $V_+ = V_-$; (3) the output drives any required current. For Rule 2: in a linear circuit the output is bounded by the rails, say $|V_{\text{out}}| \leq 3.3$ V, so $|V_+ - V_-| = |V_{\text{out}}|/A \leq 3.3/10^6 = 3.3\,\mu\text{V}$ — zero for all practical purposes. Negative feedback is what allows the circuit to settle at this equilibrium: the output moves in the direction that drives the input difference toward zero. Without feedback (or with the output saturated) the rule fails, e.g., a bare op-amp with 0.5 V across its inputs would "want" to output $A \times 0.5$ V and instead slams into the rail.

Solution 38. (a) Gain $= -47/4.7 = -10$; $V_{\text{out}} = -10 \times (-0.120) = +1.20$ V. (b) The $-$ node is a virtual ground, so $I = V_{\text{in}}/R_i = -0.120/4700 = -25.5\,\mu\text{A}$ (i.e., $25.5\,\mu\text{A}$ flowing out of the source node toward the input). (c) $R_f = 33 \times 4.7\,\text{k}\Omega = 155\,\text{k}\Omega$ (nearest standard value $150\,\text{k}\Omega$ or $160\,\text{k}\Omega$).

Solution 39. (a) Required gain $= 3.0/0.060 = 50$. Non-inverting gain $1 + R_f/R_i = 50 \Rightarrow R_f = 49\,\text{k}\Omega$ with $R_i = 1\,\text{k}\Omega$ (use $49.9\,\text{k}\Omega$, a standard 1% value). (b) An inverting stage would map 0 to +60 mV onto 0 to -3 V, i.e., outputs *below* ground — unreachable on a single 0/3.3 V supply without an added bias/reference network. (c) With $R_f \to \infty$ no feedback current flows; the divider feedback fraction goes to zero... more precisely the gain $1+R_f/R_i \to \infty$: the amplifier is open-loop, and any positive input drives the output to the positive rail (a comparator, not an amplifier).

Solution 40. (a) $R_f = 3.0\,\text{V}/2.5\,\mu\text{A} = 1.2\,\text{M}\Omega$. (b) Total current spans 0.8 to $3.3\,\mu\text{A}$, so $|V_{\text{out}}|$ spans 0.96 to 3.96 V — the top of the range exceeds the 3.3 V rail: the amplifier clips and the PPG peaks are flattened. (c) In the oximeter read circuit, Stage 1 converts the photocurrent (AC + DC) to a voltage; a DAC then injects/subtracts the measured DC component before Stage 2, so Stage 2 amplifies *only* the small AC part. Removing the 0.8-μA pedestal both prevents clipping and lets the AC use the full ADC span (larger effective resolution on the physiological signal).

Solutions G. Signal Processing

Solution 41. The sine's spectrum is a single line at f. The square wave contains the odd harmonics $f, 3f, 5f, \ldots$ with amplitudes $\propto 1/k$ (specifically $4/(\pi k)$); even harmonics are absent by symmetry. Low-pass filtering just above f removes everything but the fundamental, so the output is (nearly) a pure sine of amplitude $4/\pi \approx 1.27$ times the square wave's amplitude — filtering a square wave is a standard way to *make* a sine.

Solution 42. (a) Nyquist $= f_s/2 = 25$ Hz. (b) $84/60 = 1.4$ Hz; harmonics at $1.4k \leq 10 \Rightarrow k \leq 7.1$, so 7 harmonics fit. (c) A *notch* (band-stop) filter at 50 Hz, implemented in analog hardware before the ADC. It cannot be done afterward because 50 Hz is above the 25 Hz Nyquist limit: the interference aliases down to $|50 - f_s| = 0$ Hz — it folds onto DC/low frequencies on top of the signal band, where no filter can separate it.

Solution 43. (a) Intervals: $0.82, 0.85, 0.81, 0.85, 0.83$ s $\rightarrow 73.2, 70.6, 74.1, 70.6, 72.3$ BPM; mean interval 0.832 s $\rightarrow$ mean HR $= 60/0.832 = 72.1$ BPM. (b) The 1.22 Hz peak is the heart rate: $60 \times 1.22 = 73.2$ BPM (consistent with (a)). The global maximum at 0.083 Hz (12 s period) is residual baseline drift/respiratory-type wander — exactly why the spectral search must be restricted to the physiologic band.

Solution 44.

```python
import numpy as np
from scipy.signal import find_peaks

fs = 100.0; t = np.arange(0, 10, 1/fs); f0 = 1.5      # 90 BPM
rng = np.random.default_rng(3)
raw = (np.sin(2*np.pi*f0*t) + 0.3*np.sin(2*np.pi*2*f0*t)
       + 0.8*np.sin(2*np.pi*0.08*t) + 0.2*rng.standard_normal(t.size))

mov   = lambda x, w: np.convolve(x, np.ones(w)/w, mode="same")
sm    = mov(raw, 5)                          # smoothing
corr  = sm - mov(sm, int(2*fs) | 1)          # baseline subtraction
edge  = int(fs)
pk, _ = find_peaks(corr[edge:-edge], distance=int(0.4*fs),
                   prominence=0.5)
pk   += edge
hr_bb = (60.0 / np.diff(t[pk])).mean()

seg  = corr[edge:-edge] - corr[edge:-edge].mean()
X, f = np.fft.rfft(seg), np.fft.rfftfreq(seg.size, 1/fs)
band = (f > 0.5) & (f < 4.0)
hr_f = 60.0 * f[band][np.argmax(np.abs(X)[band])]
print(hr_bb, hr_f)      # -> 90.6 and 90.0 BPM
```

Both estimates land within a fraction of a BPM of the 90-BPM ground truth; the beat-to-beat estimate scatters slightly (noise moves individual peaks) while the FFT estimate is quantized to the spectral bin width $f_s/N \approx 0.0125$ Hz $= 0.75$ BPM.

Solution 45.

```python
from scipy.signal import butter, filtfilt, find_peaks
b, a = butter(3, (0.5, 10.0), btype="bandpass", fs=fs)
filt = filtfilt(b, a, raw)              # zero-phase filtering
```

```
pk2, _ = find_peaks(filt, distance=int(0.4*fs), prominence=0.5)
print((60.0 / np.diff(t[pk2])).mean())   # -> ~90 BPM again
```

A band-pass passes only 0.5–10 Hz: its low edge blocks the 0.08 Hz drift (the job of baseline subtraction) and its high edge blocks the broadband noise above the PPG harmonics (the job of smoothing). Both earlier steps were crude filters — the moving average a low-pass, the subtracted slow moving average a high-pass — so one properly designed band-pass replaces the pair. `filtfilt` applies the filter forward and backward for zero phase shift, so peak *times* are not displaced — essential when HR comes from peak intervals.